RK3576 System Suspend Configuration Guide

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Preface

Overview

This document provides guidance on configuring the RK3576 System suspend mode to meet product requirements.

Product Version

Chip Models	Kernel Version
RK3576	6.1

Target Audience

This document (this guide) is primarily intended for the following engineers:

Technical Support Engineers

Software Development Engineers

Revision History

Version	Author	Date	Modification Description
V1.0.0	Huang Xiaodong	2024-05-08	Initial Version

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1. System Suspend

For SoC platforms that include BL31, the system suspend operations are completed within BL31. Since the implementation of System suspend varies across different platforms' BL31, there is **no correlation or reference between standby configuration options/methods of different platforms; this document is solely applicable to the RK3576 platform.**

The general process for System suspend typically involves the following actions: shutting down the power domain, module IPs, clocks, PLLs, entering self-refresh for DDR, switching the system bus to a low-speed clock (24M or 32K), powering off vdd_arm/vdd_logic, and configuring wake-up sources. To meet the standby mode requirements of different products, the relevant configurations are currently passed to BL31 through DTS nodes during the boot phase.

1.1 Driver Files

```
./drivers/soc/rockchip/rockchip_pm_config.c
./drivers/firmware/rockchip_sip.c
./include/dt-bindings/suspend/rockchip-rk3576.h
```

1.2 DTS Nodes

```
rockchip_suspend: rockchip-suspend {
    compatible = "rockchip,pm-config";
    status = "okay";
    // Sleep log switch configuration, 0: Disable printing, 1: Enable printing
    rockchip,sleep-debug-en = <1>;
    // General configuration
    rockchip,sleep-mode-config = <
        (0
        | RKPM_SLP_ARMOFF_LOGOFF
        | RKPM_SLP_PMU_PMUALIVE_32K
        | RKPM_SLP_PMU_DIS_OSC
        | RKPM_SLP_PMIC_LP
        | RKPM_SLP_32K_EXT
        )
    >;
    // Wake-up source configuration
    rockchip,wakeup-config = <
        (0
        | RKPM_GPIO_WKUP_EN
        )
    >;
    // IO retention function configuration
```

```
    rockchip,sleep-io-ret-config = <
        (0)
    >;
    // Sleep pin configuration
    rockchip,sleep-pin-config = <
        (0
        | RKPM_SLEEP_PIN0_EN
        )
        (0)
    >;
};
```

2. DTS Configuration

Currently supported configuration options are defined in:

```
./include/dt-bindings/suspend/rockchip-rk3576.h
```

2.1 General Configuration

Attributes

```
rockchip,sleep-mode-config = <...>;
```

Configuration Items

```
// Power off vdd_arm, and the DDR controller is powered off, hardware circuit design must
support
#define RKPM_SLP_ARMOFF_DDRPD       BIT(2)
// Power off vdd_arm and vdd_log, hardware circuit design must support
#define RKPM_SLP_ARMOFF_LOGOFF      BIT(3)
// Power off vdd_arm and vdd_log, and PMU1 power domain is powered off, hardware circuit
design must support
#define RKPM_SLP_ARMOFF_PMUOFF      BIT(4)

// Use 32K clock source as system clock during sleep
#define RKPM_SLP_PMU_PMUALIVE_32K   BIT(9)
// Disable 24M crystal oscillator, can be enabled in the lowest power consumption mode,
must be used in conjunction with RKPM_SLP_PMU_PMUALIVE_32K
#define RKPM_SLP_PMU_DIS_OSC        BIT(10)
// Whether to use an external 32K clock source during sleep, if this option is not
configured, the internal 32K clock source is used by default, must be used in conjunction
with RKPM_SLP_PMU_PMUALIVE_32K
#define RKPM_SLP_32K_EXT            BIT(24)
// Sleep timer wake-up function, for test use
#define RKPM_SLP_TIME_OUT_WKUP      BIT(25)
```

```
// Sleep output PMU state machine waveform, for debug use
#define RKPM_SLP_PMU_DBG            BIT(26)
```

Notes

- Configure the sleep settings according to the product's wake-up source requirements. For example, if USB wake-up is required, the USB power domain and clocks must remain enabled during sleep. Therefore, options such as RKPM_SLP_ARMOFF_LOGOFF, RKPM_SLP_PMU_DIS_OSC, and RKPM_SLP_PMU_PMUALIVE_32K should not be enabled.
- RKPM_SLP_PMU_DBG: When this configuration is enabled, the PMU state machine will output a specific waveform signal through GPIO0_B0 during standby, which is used to feedback the internal state of the PMU state machine. This function is only used for sleep wake-up testing and debug.
- RKPM_SLP_TIME_OUT_WKUP: When this configuration is enabled, the system will automatically wake up after about 1 second during standby, and only the PMU internal timer can wake up the system. This configuration is only used for sleep wake-up testing and debug.

2.2 Wake-Up Configuration

Attributes

```
rockchip,wakeup-config = <...>;
```

Configuration Items

```
// Supports all interrupt wake-ups (sleep-wake interrupts managed by GIC), generally the
sleep process is run by cpu0, and only RKPM_CPU0_WKUP_EN needs to be configured when needed
#define RKPM_CPU0_WKUP_EN       BIT(0)
#define RKPM_CPU1_WKUP_EN       BIT(1)
#define RKPM_CPU2_WKUP_EN       BIT(2)
#define RKPM_CPU3_WKUP_EN       BIT(3)
#define RKPM_CPU4_WKUP_EN       BIT(4)
#define RKPM_CPU5_WKUP_EN       BIT(5)
#define RKPM_CPU6_WKUP_EN       BIT(6)
#define RKPM_CPU7_WKUP_EN       BIT(7)
// GPIO0 wake-up
#define RKPM_GPIO_WKUP_EN       BIT(8)
// SDMMC wake-up
#define RKPM_SDMMC_WKUP_EN      BIT(9)
// SDIO wake-up
#define RKPM_SDIO_WKUP_EN       BIT(10)
// USB DEV wake-up
#define RKPM_USB_WKUP_EN        BIT(11)
// UART1 wake-up
#define RKPM_UART_WKUP_EN       BIT(12)
// MCU wake-up
#define RKPM_MCU_WKUP_EN        BIT(13)
// PMU TIMER wake-up
#define RKPM_TIMER_WKUP_EN      BIT(14)
// Supports all interrupt wake-ups (not managed by GIC), not recommended for use
```

```
#define RKPM_SYSINT_WKUP_EN     BIT(15)
// pwm wake-up, generally used for infrared wake-up
#define RKPM_PWM_WKUP_EN        BIT(16)
// tsadc wake-up
#define RKPM_TSADC_WKUP_EN      BIT(17)
// hptimer wake-up
#define RKPM_HPTIMER_WKUP_EN    BIT(18)
// saradc wake-up
#define RKPM_SARADC_WKUP_EN     BIT(19)
// PMU internal timer wake-up (default 1s), used for testing and debugging sleep-wake
#define RKPM_TIMEOUT_WKUP_EN    BIT(20)
```

Notes

- RKPM_GPIO_WKUP_EN (Preferred):

  Only GPIO0 among GPIO0~4 supports the pin group as a wake-up source. In this mode, the interrupt signal on the pin of GPIO0 is sent directly to the PMU state machine without going through GIC. In hardware design, it is recommended that users place all required wake-up sources on the GPIO0 pin group.

- RKPM_CPU0_WKUP_EN (Secondary):

  Supports all wake-up interrupts registered with GIC using enable_irq_wake() during the kernel phase, offering more wake-up interrupt sources than RKPM_GPIO_WKUP_EN. However, this method decentralizes the management of wake-up sources to various kernel modules, and the system may be awakened by unexpected interrupts during standby.

- RKPM_TIMEOUT_WKUP_EN:

  The PMU's internal timer wake-up, with a default 1s timeout to generate an interrupt, is generally used only for testing sleep-wake during the development phase.

2.3 IO Retention Configuration

In sleep scenarios where vdd_logic is powered off, if users wish to maintain the voltage level of a specific IO, they can configure this attribute.

Attributes

```
rockchip,sleep-io-ret-config = <
    (0
    | RKPM_VCCIO0_RET_EN
    ...
    )
>;
```

Configuration Items

```
// The following nine IO domains can be configured
#define RKPM_VCCIO0_RET_EN              BIT(0)
#define RKPM_VCCIO1_RET_EN              BIT(1)
#define RKPM_VCCIO2_RET_EN              BIT(2)
#define RKPM_VCCIO3_RET_EN              BIT(3)
#define RKPM_VCCIO4_RET_EN              BIT(4)
#define RKPM_VCCIO5_RET_EN              BIT(5)
#define RKPM_VCCIO6_RET_EN              BIT(6)
#define RKPM_VCCIO7_RET_EN              BIT(7)
#define RKPM_PMUIO1_RET_EN              BIT(8)
```

Notes

- To maintain the voltage level of a specific IO, users must configure the IO domain where this IO is located, and the power supply of this IO domain must not be powered off; otherwise, it is impossible to maintain the voltage level.

2.4 Sleep Pin Configuration

The sleep pin is used to cut off power during sleep mode, and users need to configure it according to the actual hardware design.

Attributes

```
rockchip,sleep-pin-config = <
        (0
        | RKPM_SLEEP_PIN0_EN
        ...
        )
        (0
        | RKPM_SLEEP_PIN0_ACT_LOW
        ...
        )
>;
```

Configuration Items

```
// The following three sleep pins are supported
#define RKPM_SLEEP_PIN0_EN                      BIT(0)  /* GPIO0_A3 */
#define RKPM_SLEEP_PIN1_EN                      BIT(1)  /* GPIO0_A4 */
#define RKPM_SLEEP_PIN2_EN                      BIT(2)  /* GPIO0_A5 */

// If pulling down the sleep pin to cut off power, configure the corresponding item below.
Default is to pull up the sleep pin.
#define RKPM_SLEEP_PIN0_ACT_LOW         BIT(0)  /* GPIO0_A3 */
#define RKPM_SLEEP_PIN1_ACT_LOW         BIT(1)  /* GPIO0_A4 */
#define RKPM_SLEEP_PIN2_ACT_LOW         BIT(2)  /* GPIO0_A5 */
```

Notes

- The default configuration for RK3576 is to pull up the sleep pin0.

2.5 Virtual Power-off Configuration

Some products cannot power off when shutting down, requiring the use of a sleep scenario to replace the power-off scenario to reduce power consumption.

Attributes

```
// 0: Disable virtual power-off mode, 1: Enable virtual power-off mode.
rockchip,virtual-poweroff = <...>;

// Configure the wake-up interrupt number when shutting down, the system restarts upon
receiving the wake-up interrupt signal
rockchip,virtual-poweroff-irqs = < xx xx xx ...>;
```

Notes

- If rockchip,virtual-poweroff-irqs is not configured, pwm0 channel0 will be used as the wake-up source by default.

3. Print Information

The following section provides a brief description of the BL31 log messages generated during system suspend and wake-up. For clarity, some log messages are shown on separate lines in the examples below. Different sleep modes may produce different log outputs. Refer to the actual logs displayed on the target system for the final results.

Attributes

```
// 0: No log will be printed during sleep, 1: Log will be printed during sleep
rockchip,sleep-debug-en = <...>;
```

RK3576 System Suspend Print

```
// The version and commit information of the BL31 used for sleep
INFO:    BL31: v2.3():v2.3-775-gacfcb4fbd:derrick.huang, fwver: v1.05

// Sleep mode configuration and sleep count print
INFO:    enter: cfg=0x1020608, sleeptimes:1

// Sleep mode print
INFO:    deep
INFO:    armoff_logoff
INFO:    pmu_pmualive_32k
INFO:    pmu_dis_osc
INFO:    pmic_low_power
```

```
INFO:    32k ext
INFO:    io_ret (0x8)
INFO:    sleep_pin: 0x1 0x0
INFO:    GPIO POWER INFO:
INFO:           not config

// Sleep gpio interrupt configuration status print
INFO:    GPIO0_INTEN: 0xffff 0xffff 0xfd9e 0xffff 0x0 0xd408611e
INFO:    GPIO1_INTEN: 0xffff 0xffff 0xffff 0xffff 0x0 0xef03f600
INFO:    GPIO2_INTEN: 0xffff 0xffff 0xffff 0xffff 0x0 0x9fdb0c20
INFO:    GPIO3_INTEN: 0xffff 0xffff 0xffff 0xffff 0x0 0xaf8c3ec4
INFO:    GPIO4_INTEN: 0xffff 0xffff 0xffff 0xffff 0x0 0xfe92c7d7

// Sleep system interrupt status print
INFO:     IRQ_EN: 77
INFO:     IRQ_EN: 108
INFO:     IRQ_EN: 135
INFO:     IRQ_EN: 132
INFO:     IRQ_EN: 189
INFO:     IRQ_EN: 185
INFO:     IRQ_EN: 201
INFO:     IRQ_EN: 197
INFO:     IRQ_EN: 193
INFO:     IRQ_EN: 312
INFO:     IRQ_PED: 268

// Sleep key register information print
INFO:    PMU1_PWR_CON(0x2101) PMU1_CRU_PWR_CON(0x22f) PMU1_WAKEUP_INT_CON(0x100)
PMU2_BUS_IDLE_ST(0x7ffff) PMU2_PWR_GATE_ST(0x3ffffc1)
PMU2_BUS_IDLE_CON(0x0 0x3ff8)
PMU2_PWR_GATE_CON(0x3e 0x0)
PMU2_VOL_GATE_CON(0x31 0x200)
PMU1_DDR_PWR_CON(0x2b 0x2b)
PMU1_PLLPD_CON(0xfff)
INFO:    PMU0_PWR_CON(0x0) PMU0_WAKEUP_INT_CON(0x0)
PMU0_DDR_RET_CON(0x0 0x0) PMU0_GRF_OS_REGS17(0xe707d445)
PMU0GRF_SOC_CON5(0x64)
```

RK3576 System Wakeup Print

```
// Wakeup process step print
012abcdef3456789
987654fedcba3210

// Wakeup source
INFO:    wake up status: 0x100
INFO:    GPIO0 interrupt wakeup: 0x40

// If gpio wakes up, it will print the specific gpio
INFO:      gpio0_a6
```

4. Typical Configuration Examples

4.1 Logic Power-Off Mode

This mode supports the power-off of vdd_logic, reducing sleep power consumption.

Configuration as follows:

```
	rockchip_suspend: rockchip-suspend {
		compatible = "rockchip,pm-config";
		status = "okay";
		rockchip,sleep-debug-en = <1>;
		rockchip,sleep-mode-config = <
			(0
			| RKPM_SLP_ARMOFF_LOGOFF
			| RKPM_SLP_PMU_PMUALIVE_32K
			| RKPM_SLP_PMU_DIS_OSC
			| RKPM_SLP_PMIC_LP
			| RKPM_SLP_32K_EXT
			)
		>;
		rockchip,wakeup-config = <
			(0
			| RKPM_GPIO_WKUP_EN
			)
		>;
		rockchip,sleep-io-ret-config = <
			(0)
		>;
		rockchip,sleep-pin-config = <
			(0
			| RKPM_SLEEP_PIN0_EN
			)
			(0)
		>;
	};
```

4.2 PMU1 Power-off Mode

This mode, based on the logic power-off, turns off PD PMU1 and supports the shutdown of PMUIO1 power supply, with lower power consumption than the logic power-off mode.

Configuration as follows:

```
	rockchip_suspend: rockchip-suspend {
		compatible = "rockchip,pm-config";
		status = "okay";
		rockchip,sleep-debug-en = <1>;
```

```
                rockchip,sleep-mode-config = <
                        (0
                        | RKPM_SLP_ARMOFF_PMUOFF
                        | RKPM_SLP_PMU_PMUALIVE_32K
                        | RKPM_SLP_PMU_DIS_OSC
                        | RKPM_SLP_PMIC_LP
                        | RKPM_SLP_32K_EXT
                        )
                >;
                rockchip,wakeup-config = <
                        (0
                        | RKPM_GPIO_WKUP_EN
                        )
                >;
                rockchip,sleep-io-ret-config = <
                        (0)
                >;
                rockchip,sleep-pin-config = <
                        (0
                        | RKPM_SLEEP_PIN0_EN
                        )
                        (0)
                >;
        };
```

4.3 UART1 Wakeup

If the product requires UART wakeup functionality, it is advisable to opt for UART1 whenever possible. This is because UART1 is located in the PD_PMU1 and can support UART1 wakeup even when the logic power is off, resulting in lower power consumption.

Configuration as follows:

```
        rockchip_suspend: rockchip-suspend {
                compatible = "rockchip,pm-config";
                status = "okay";
                rockchip,sleep-debug-en = <1>;
                rockchip,sleep-mode-config = <
                        (0
                        | RKPM_SLP_ARMOFF_LOGOFF
                        | RKPM_SLP_PMIC_LP
                        )
                >;
                rockchip,wakeup-config = <
                        (0
                        | RKPM_GPIO_WKUP_EN
                        | RKPM_UART_WKUP_EN
                        )
                >;
                rockchip,sleep-io-ret-config = <
                        (0)
```

```
                >;
                rockchip,sleep-pin-config = <
                        (0
                        | RKPM_SLEEP_PIN0_EN
                        )
                        (0)
                >;
        };
```

Notes

- The UART1 wakeup requires the 24M crystal oscillator to remain active, hence the power supply to the 24M crystal oscillator must not be turned off.

4.4 USB Wakeup

If the product requires USB wakeup functionality, it is advisable to merge the USB power supplies of 0.75V, 1.8V, and 3.3V with the PMU's power supply. This allows for USB wakeup even when the logic power is cut off, resulting in lower power consumption.

Configuration as follows:

```
        rockchip_suspend: rockchip-suspend {
                compatible = "rockchip,pm-config";
                status = "okay";
                rockchip,sleep-debug-en = <1>;
                rockchip,sleep-mode-config = <
                        (0
                        | RKPM_SLP_ARMOFF_LOGOFF
                        | RKPM_SLP_PMIC_LP
                        )
                >;
                rockchip,wakeup-config = <
                        (0
                        | RKPM_GPIO_WKUP_EN
                        | RKPM_USB_WKUP_EN
                        )
                >;
                rockchip,sleep-io-ret-config = <
                        (0)
                >;
                rockchip,sleep-pin-config = <
                        (0
                        | RKPM_SLEEP_PIN0_EN
                        )
                        (0)
                >;
        };
```

Notes

- USB wakeup requires the 24M crystal oscillator to remain active, hence the power supply to the 24M crystal oscillator must not be turned off.

4.5 PWM Wakeup

If the product requires infrared wake-up functionality, PWM wake-up should be utilized, and it is advisable to connect the infrared to PWM0 channel0. Since PWM0 is in PD_PMU1, it can support PWM wake-up even when the logic power is off, resulting in lower power consumption.

Configuration as follows:

```
	rockchip_suspend: rockchip-suspend {
		compatible = "rockchip,pm-config";
		status = "okay";
		rockchip,sleep-debug-en = <1>;
		rockchip,sleep-mode-config = <
			(0
			| RKPM_SLP_ARMOFF_LOGOFF
			| RKPM_SLP_PMIC_LP
			)
		>;
		rockchip,wakeup-config = <
			(0
			| RKPM_GPIO_WKUP_EN
			| RKPM_PWM_WKUP_EN
			)
		>;
		rockchip,sleep-io-ret-config = <
			(0)
		>;
		rockchip,sleep-pin-config = <
			(0
			| RKPM_SLEEP_PIN0_EN
			)
			(0)
		>;
	};
```

Notes

- PWM wake-up requires the 24M crystal oscillator to remain active, hence the power supply to the 24M crystal oscillator must not be turned off.

4.6 MCU Wakeup

If the product requires the PMU MCU to perform some tasks during sleep (such as interrupt filtering, etc.), then MCU wakeup is necessary.

Configuration as follows:

```
	rockchip_suspend: rockchip-suspend {
		compatible = "rockchip,pm-config";
		status = "okay";
		rockchip,sleep-debug-en = <1>;
		rockchip,sleep-mode-config = <
			(0
			| RKPM_SLP_ARMOFF_DDRPD
			| RKPM_SLP_PMIC_LP
			)
		>;
		rockchip,wakeup-config = <
			(0
			| RKPM_GPIO_WKUP_EN
			| RKPM_MCU_WKUP_EN
			)
		>;
		rockchip,sleep-io-ret-config = <
			(0)
		>;
		rockchip,sleep-pin-config = <
			(0
			| RKPM_SLEEP_PIN0_EN
			)
			(0)
		>;
	};
```

Notes

- MCU wakeup requires the 24M crystal oscillator to remain active, so the power supply to the 24M crystal oscillator must not be turned off.
- VDD_LOGIC cannot be turned off for MCU wakeup.

4.7 GPIO Wakeup

If the product requires wakeup via GPIO0, since GPIO0 is within the PD_PMU, the configuration for the logic power-off mode can be used.

For wakeup via GPIO1 to 4, VDD_LOGIC must not be powered off, and the configuration is as follows:

```
	rockchip_suspend: rockchip-suspend {
		compatible = "rockchip,pm-config";
		status = "okay";
		rockchip,sleep-debug-en = <0>;
		rockchip,sleep-mode-config = <
			(0
			| RKPM_SLP_ARMOFF_DDRPD
			| RKPM_SLP_PMIC_LP
			)
		>;
		rockchip,wakeup-config = <
```

```
                        (0
                        | RKPM_GPIO_WKUP_EN
                        | RKPM_CPU0_WKUP_EN
                        )
                >;
                rockchip,sleep-io-ret-config = <
                        (0)
                >;
                rockchip,sleep-pin-config = <
                        (0
                        | RKPM_SLEEP_PIN0_EN
                        )
                        (0)
                >;
        };
```

Notes

- Wakeup via GPIO1 to 4 requires the 24M crystal oscillator to remain active, hence the power supply to the 24M crystal oscillator must not be turned off.